Repeated games

Will it end?

Play
Bid High or Low. Then roll the die.

→

Shorten
Carry on only for a 5 or 6. Play again.

→

Explain
Why the future changes today.

Podium:

	High	Low
High	3	0
Low	5	1

Your payoff as the row bidder.

Played once, what should you do?

Played once, what do you both get?

And if you had both bid High?

Low is a dominant action in the one-shot game, which is exactly the problem.

We stop on a 1, so the game carries on with probability $\delta =$ ____.

The expected number of rounds is $\frac{1}{1-\delta} =$ ____.

Now we carry on *only* for a 5 or a 6, so $\delta =$ ____ and the expected number of rounds is ____.

Did the shorter game change how you played? ____

Grudger plays High until you play Low, then Low for ever. Hold your opponent to it.

You conform, and play High for ever

$3 + 3\delta + 3\delta^2 + \cdots =$ ☐

You deviate to Low in round one

$5 + \delta + \delta^2 + \cdots =$ ☐

So cooperating is worth it when ☐ $\geq$ ☐.

$$\frac{3}{1-\delta} \geq 5 + \frac{\delta}{1-\delta} \quad \Longleftrightarrow \quad \delta \geq \square$$

Our first die gives $\delta = 5/6$, so cooperation ☐ be sustained. The second gives $\delta = 1/3$, so it ☐.

Which side of the line did you guess before we played? ☐

In a round you know is the last, the future is worth ____, so you should bid ____.

Both of you know that, so in the round before it the future is also worth nothing, and you bid ____. Keep going backwards and cooperation _____.

So the die was doing real work: not by making the game long, but by making the end _____.

What we just did in the room is **Question 1** on the Repeated Games page.

(a)	Interpret $\delta = 5/6$ and give the expected number of rounds.	[2]
(b)	Discounted payoffs against Grudger, conforming and deviating.	[6]
(c)	Is cooperation sustained at $\delta = 5/6$?	[4]
(d)	The smallest δ that sustains it, and what our die meant.	[5]
(e)	The same for a forgiving trigger that punishes for two rounds only.	[7]

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